

Internal-Flow CFD Analysis

Elbows, a Conical Diffuser, and a Converging–Diverging Nozzle

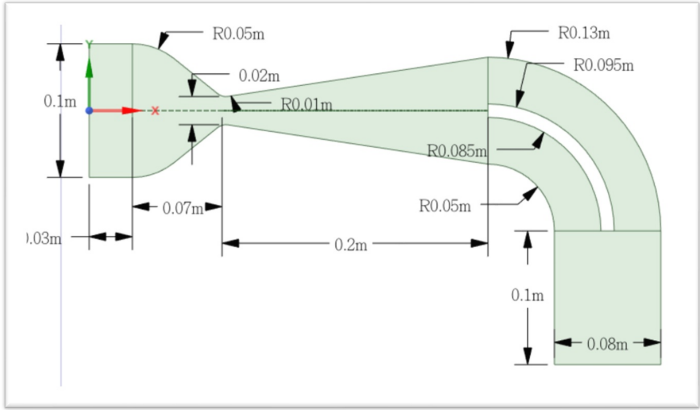


Figure – Computational domain of the converging–diverging nozzle with elbow and guide vanes.

PROJECT	Numerical Investigation of Internal Flows and Local Losses
SOFTWARE	ANSYS Fluent and MATLAB post-processing
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COURSE	Fluid Mechanics 1 – Dr. Karimi – Spring 1404
REPORT	English Portfolio Edition – 2025

1. Project Overview

This project studies three internal-flow geometries of increasing complexity. The first phase compares two 90-degree elbows with different curvature ratios. The second examines a 40-degree conical diffuser and quantifies pressure recovery and separation. The third investigates a non-isothermal converging–diverging nozzle connected to an elbow and a guide-vane outlet.

Study matrix

Phase	Geometry	Fluid and condition	Primary outputs
1	90° elbow, R/D = 2 and 4	Water, Re = 500 and 10,000	Pressure loss, velocity profiles, separation
2	40° conical diffuser	Water, inlet velocity ≈ 0.125 m/s	Pressure recovery, loss coefficient, recirculation
3	Elbow, guide vane, CD nozzle	Air, Re ≈ 5,500, inlet at 600 K	Velocity, pressure, temperature, density, mass balance

Project objectives

- Evaluate how curvature ratio changes separation and local pressure loss in a pipe elbow.
- Quantify flow separation and incomplete pressure recovery in a large-angle diffuser.
- Track pressure, velocity, temperature and density through a coupled nozzle system.
- Check mesh independence and mass conservation before interpreting the final fields.
- Use MATLAB to reconstruct contours, profiles and streamlines from exported Fluent data.

2. Numerical Method and Evaluation Metrics

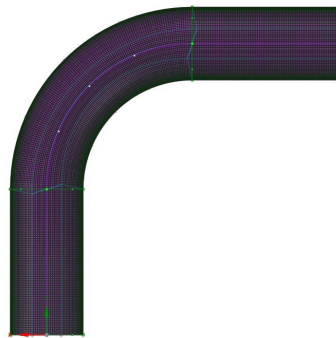
The simulations use finite-volume discretization in ANSYS Fluent. Velocity-inlet and pressure-outlet boundary conditions are applied, with no-slip conditions on solid walls. Turbulence (standard or RNG k- ϵ), energy transport and variable density are activated where required. Calculations continue until residuals and monitored engineering quantities stabilize.

$K = \Delta p / (\frac{1}{2} \rho V^2)$ $Re = \rho V D / \mu$ For the diffuser, recovery coefficient $Cr = -\xi$ when pressure rises.

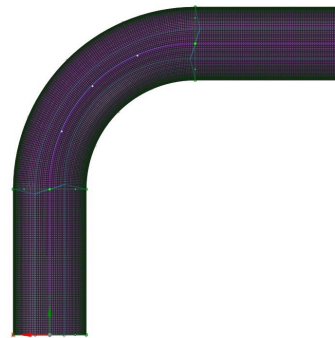
Check	Reported criterion	Purpose
Residuals	~ 10 ⁻⁴	Iterative convergence
Mesh (elbow)	Orth. Quality > 0.69; AR < 4.4; ~662k cells	Spatial discretization quality
Mesh (diffuser)	Orth. Quality > 0.76; AR < 4.4; ~163k cells	Spatial discretization quality
Mass balance	Imbalance ≈ 1.7 g/s (Phase 3)	Verify conservation

3. Phase 1 – Flow Through a 90° Elbow

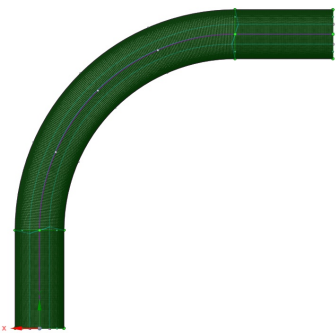
Two centerline curvature ratios, $R/D = 2$ and $R/D = 4$, are studied at Reynolds numbers 500 (laminar) and 10 000 (turbulent). Pipe diameter is 0.1 m. Water is the working fluid ($\rho = 1000 \text{ kg/m}^3$, $\mu = 0.001 \text{ Pa}\cdot\text{s}$). Inlet velocities are set to achieve the target Reynolds numbers (0.005 m/s for $Re = 500$ and 0.1 m/s for $Re = 10\,000$).



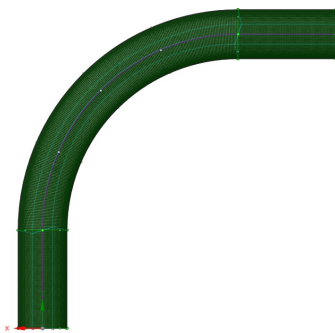
Elbow $R/D = 2$ (mesh)



Elbow $R/D = 2$ (mesh, alternate view)



Elbow $R/D = 4$ (mesh)

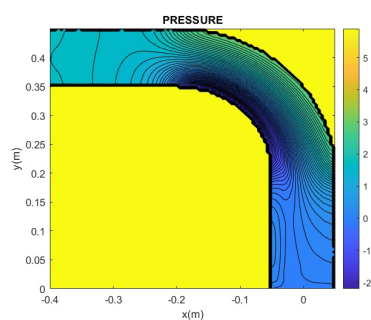


Elbow $R/D = 4$ (mesh, alternate view)

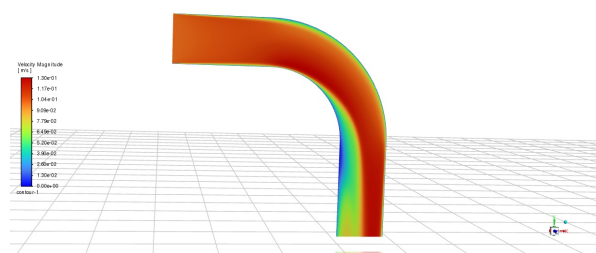
The sharper elbow ($R/D = 2$) produces stronger cross-stream motion and a larger separated region near the inner wall. The wider elbow ($R/D = 4$) turns the flow more gradually, reducing separation intensity while still generating secondary flow due to centrifugal force.

Velocity profiles & pressure variation

Velocity profiles were extracted upstream of the bend, at mid-bend, and downstream. At the bend mid-section the centrifugal force drives fluid toward the outer wall. After the bend the profile gradually redevelops but remains distorted. Static pressure decreases continuously along the pipe due to wall friction and the additional local loss of the bend.



Velocity profile (example station)



Velocity profiles comparison

- Smaller curvature ratio ($R/D = 2$) increases separation and flow distortion near the inner wall.
- Larger curvature ratio ($R/D = 4$) produces milder secondary flow and more gradual pressure change.
- Turbulent cases ($Re = 10\,000$) exhibit thinner boundary layers compared with laminar cases.

4. Phase 2 – 40° Conical Diffuser

The diffuser expands from an inlet diameter of 0.08 m to an outlet diameter of 0.16 m with a total included angle of 40°. Water enters at approximately 0.125 m/s ($Re \approx 10\,000$). A structured mesh of roughly 162 500 cells with good orthogonal quality (> 0.76) was employed. Turbulence was modelled with the RNG $k-\epsilon$ closure.

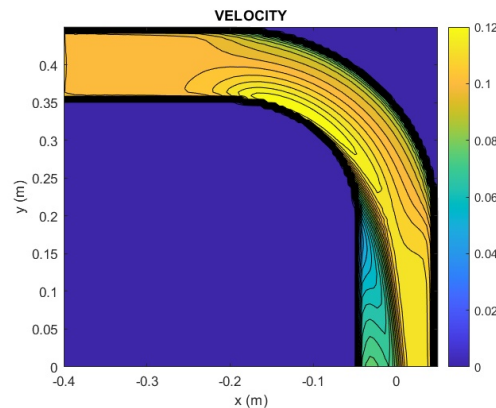
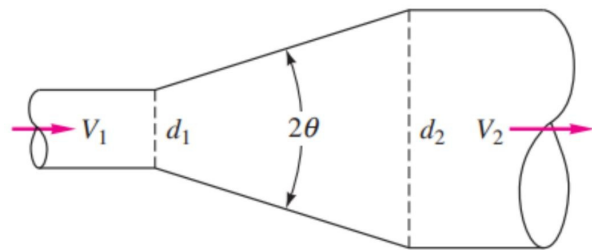
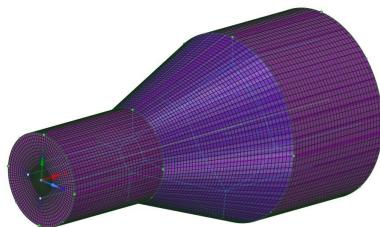


Figure – Mesh of the 40° conical diffuser (inlet $D = 0.08$ m, outlet $D = 0.16$ m).

The large expansion angle creates a strong adverse pressure gradient. A recirculating near-wall region appears while the central core remains attached. Velocity decreases through the expanding section, but pressure recovery is incomplete.



Velocity contour (MATLAB)

Pressure variation along axis

Quantity	Value	Interpretation
Inlet velocity	0.125 m/s	Reference velocity for coefficient calculation
Pressure rise Δp	≈ 0.45 Pa	Static-pressure recovery observed along the axis
Recovery coefficient Cr	0.576	Within empirical range for 40° diffusers (0.3–0.6)

According to standard hydraulic-resistance data (Idelchik), a 40° conical diffuser typically operates with strong separation and Cr in the range 0.3–0.6. The obtained value $Cr = 0.576$ confirms that the simulation captures the expected incomplete recovery.

5. Phase 3 – Converging–Diverging Nozzle System

The final phase combines a 90-degree elbow, a guide-vane outlet and a converging–diverging nozzle. Air enters at 0.8 m/s and 600 K; solid walls are held at 250 K. Density varies with temperature according to the ideal-gas law. The entire flow remains subsonic.

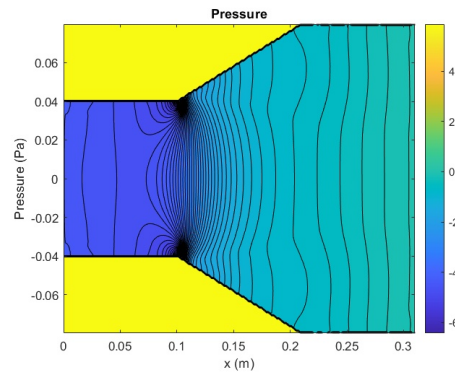
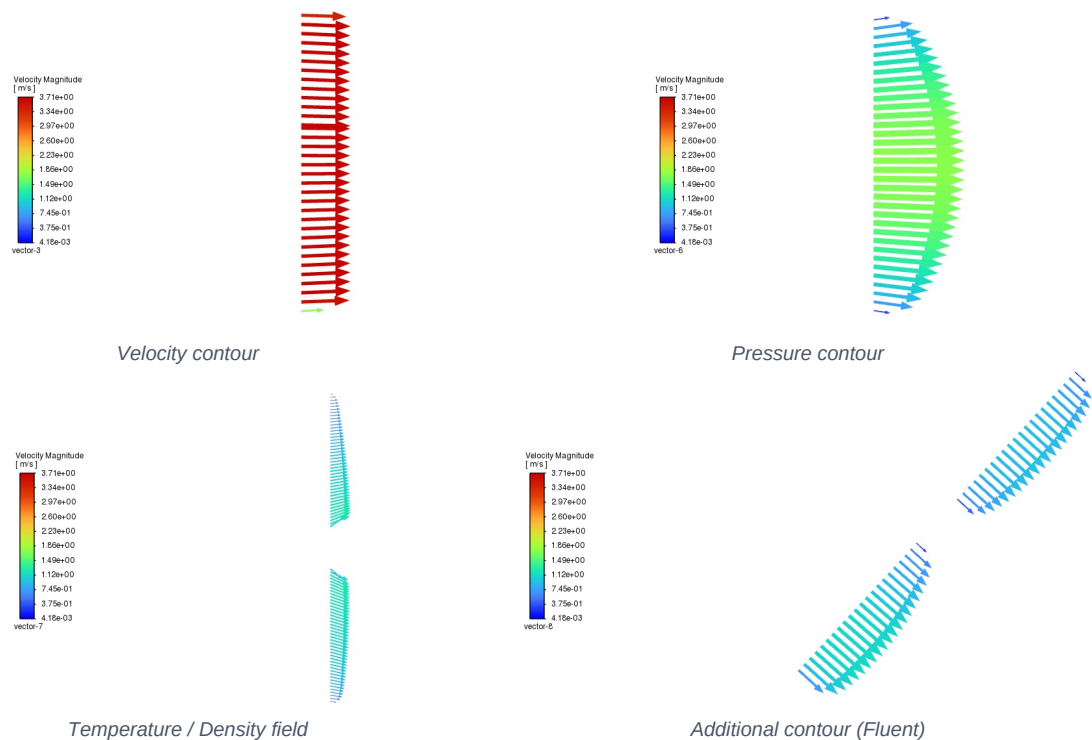


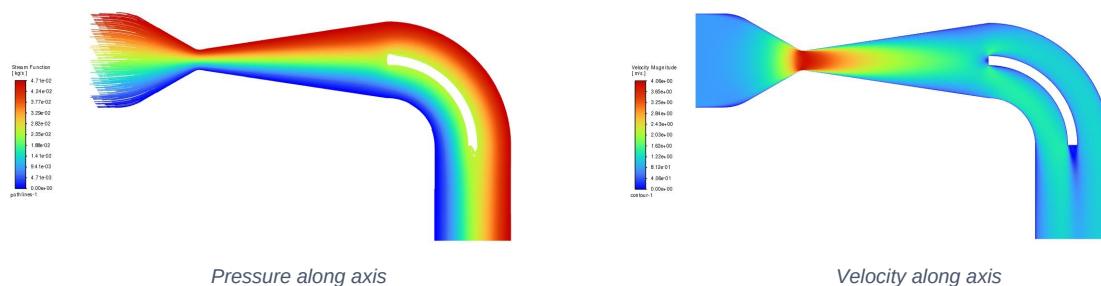
Figure – Computational domain / mesh of the nozzle–elbow–guide-vane system.

Field observations

Centerline velocity rises from 0.8 m/s to approximately 3.6 m/s near the nozzle throat and then decreases in the diverging section. Static pressure falls to about -2.5 Pa at the throat and recovers partially downstream. Core temperature remains near 480–500 K while the cooled boundary layers approach 270–300 K. Density follows the inverse of temperature and local pressure.



Centerline velocity and pressure



Mass-flow balance

Boundary	Mass-flow rate (kg/s)	Share of inlet
Main inlet	+0.0471	100 %
Guide-vane outlets (combined)	≈ -0.0141	$\approx 30 \%$
Main nozzle outlet	≈ -0.0347	$\approx 74 \%$

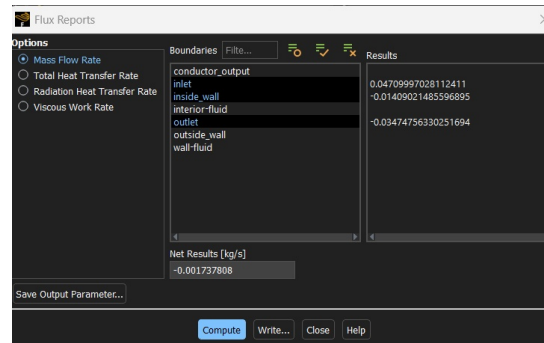


Figure – Mass-flow report extracted from ANSYS Fluent (net imbalance of order 1.7 g/s).

The absolute imbalance is of the order of 1.7 g/s (a few percent of the inlet mass flow). While larger than the ideal machine-zero residual, the discrepancy remains acceptable for the present mesh density and confirms that overall mass conservation is reasonably satisfied.

Streamlines and additional visualizations

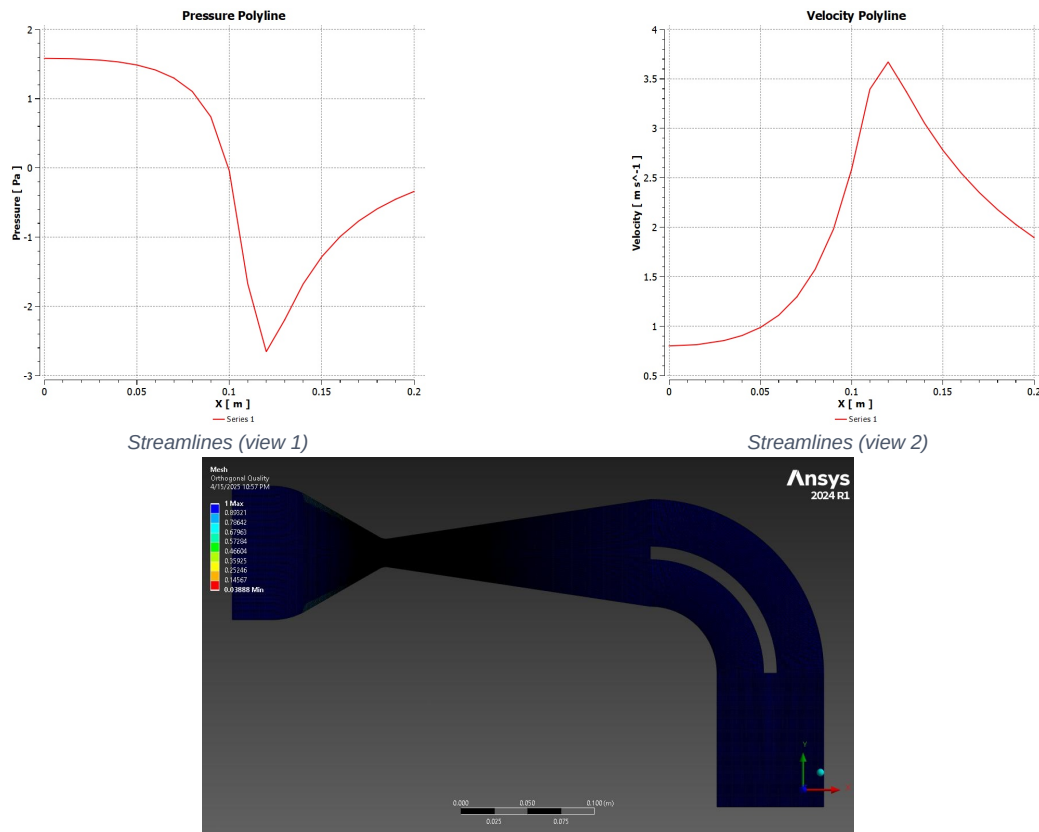


Figure – Additional flow-field visualization (velocity vectors / streamlines).

6. Conservation, Design Assessment and Conclusions

Main findings

- Smaller elbow curvature ($R/D = 2$) increases separation and flow distortion near the inner wall compared with $R/D = 4$.
- The 40° conical diffuser develops a recirculating wall region and incomplete pressure recovery; $Cr \approx 0.576$ lies inside the accepted empirical band for strong-separation diffusers.
- The final nozzle system produces strong local acceleration (peak velocity ≈ 3.6 m/s), cooling boundary layers, and density gradients driven by the cold walls.
- Mesh-quality metrics and mass-balance checks support numerical credibility.
- The guide-vane outlet extracts a non-negligible fraction of the mass flow; redesign would be needed if a precise extraction ratio is required.

Limitations

The archived project does not include complete Fluent case and mesh files, so the repository documents the analysis and post-processing workflow rather than providing a one-click re-run. Further work should preserve solver case files, apply fully developed inlet profiles, compare multiple turbulence models, and systematically benchmark local-loss coefficients against the most recent empirical correlations.

This English edition is prepared as a concise portfolio summary of the original Persian course project completed at Sharif University of Technology under the supervision of Dr. Karimi.